

Maintenance Agent

Role: Diagnostic Checker · Pipeline Position: Step 3 of 5 · Endpoint: POST
/maintenance/evaluate

Evaluates vehicle health score, schedules servicing, and alerts the driver by email.

Overview

The Maintenance Agent is the third agent in the pipeline. It reads the vehicle's health_score, applies rule-based diagnostics, inserts maintenance logs for critical vehicles, updates database status, and sends an HTML alert email to the assigned driver.

Health Score Classification

Score Range	Status	Action Taken
> 80	Healthy	Return report only
50–80	Service Recommended	Return report + next_service_after_km
< 50	Maintenance Required	Log inserted, vehicle → Maintenance, driver email s

Remaining Service Distance Formula

```
remaining_service_distance = (health_score - 50) * 50 km  
(valid when health_score >= 50)
```

API Endpoint

Method	Path	Description
POST	/maintenance/evaluate	Evaluate vehicle health and schedule maintenance if ne

Request Payload

```
{ "vehicle_id": "<uuid>" }
```

Response — Healthy

```
{ "health_score": 88, "vehicle_status": "Healthy",  
  "message": "Vehicle is healthy.", "next_service_after_km": 1900 }
```

Response — Critical

```
{ "health_score": 35, "vehicle_status": "Maintenance Required",  
  "message": "Vehicle requires immediate maintenance." }
```

Driver Email Alert (Critical Health)

- Looks up current driver via vehicles → drivers → users JOIN.
- If driver email ends in @agentfleet.com (mock domain), falls back to DEMO_DRIVER_EMAIL.
- Sends HTML email with: health score, vehicle plate, diagnostic issue, garage instructions.
- Email is sent asynchronously — pipeline continues even if SMTP fails.

Internal Logic Flow

```
evaluate_vehicle(vehicle_id)  
  
■  
■■■ 1. get_vehicle(vehicle_id) → VehicleUnavailableException if not found  
■■■ 2. get_latest_maintenance(vehicle_id) → log trace only  
■■■ 3. Rule-based health_score classification  
■■■ 4. Compute remaining_service_distance  
■■■ [if critical]:  
    ■■■ 5. insert_maintenance_log()  
    ■■■ 6. update_vehicle_health(status='Maintenance')  
    ■■■ 7. send_email_async(driver_email, subject, html_body)
```

Module Files

File	Purpose
agent.py	Agent registry entry — wraps service call
service.py	Business logic: health rules, maintenance log, email dispatch
repository.py	DB queries: vehicle fetch, log insert, health status update
routes.py	FastAPI router — POST /maintenance/evaluate + GET /maintenance
schemas.py	Pydantic request/response models

prompts.py	AI prompt templates
tools.py	LLM tool definitions